

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS1

SUBJECT: Geography

TOPIC: Weather & Climate (Revision)

Introduction

- Weather is the condition of the atmosphere at a place at a particular time; climate is the average weather of a place over a long period (30 years or more). The elements of weather are temperature, rainfall, pressure, wind, humidity and sunshine.

Elements and measurements

- **Temperature** — measured with a thermometer in a Stevenson screen (placed 1.2 m above the ground, louvered for ventilation); the daily mean = $(\text{max} + \text{min}) \div 2$.
- **Rainfall** — rain gauge: a funnel of 12.7 cm diameter over a collecting jar; water is poured into a measuring cylinder each morning.
- **Pressure** — mercury or aneroid barometer; **wind** — anemometer for speed, wind vane for direction; **humidity** — hygrometer; **sunshine** — Campbell–Stokes recorder; **clouds and evaporation** — observation and evaporimeter.

Why weather matters

- It determines agriculture (planting and harvesting), water supply, transport (visibility, flooding), health (malaria in wet areas), clothing, housing and energy use — the reason weather stations are vital.

CLASS EVALUATION

1. Define weather and climate.
2. Why is the Stevenson screen louvered?
3. How is daily rainfall measured?
4. State two ways weather affects agriculture.

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS1

SUBJECT: Geography

TOPIC: Climate Elements

Introduction

- The climate of a place is described by its temperature, rainfall, pressure, winds and humidity — the climatic elements — and their seasonal patterns.

Temperature

- Depends on latitude, altitude (falls about 6.5 °C per 1000 m), distance from the sea (inland areas have greater range), ocean currents, cloud cover and aspect. The diurnal range is the difference between day and night temperatures.

Rainfall and pressure

- Rainfall types: **convictional** (heating of the surface — the afternoon downpours of Nigeria), **orographic/relief** (air forced up mountains), **frontal/cyclonic** (warm and cold air masses meeting). Pressure belts: the equatorial low, the subtropical highs, the subpolar lows and the polar highs — they drive the planetary winds. Isobars join places of equal pressure; winds blow from high to low pressure, deflected by the earth's rotation (Coriolis effect).

Effects of ocean currents

- Warm currents (Gulf Stream, Brazil) warm and moisten coasts; cold currents (Canary, Benguela, Humboldt) cool them and bring dry weather — the reason the West African coast near the cold Canary current has lower rainfall.

CLASS EVALUATION

1. Name the three types of rainfall.
2. Explain orographic rainfall.
3. How does altitude affect temperature?
4. State one effect of a cold ocean current.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS1

SUBJECT: Geography

TOPIC: Vegetation Zones of West Africa

Introduction

- Vegetation is the plant cover of an area. It is controlled mainly by rainfall, temperature and soil; West Africa's vegetation belts run roughly parallel to the coast, reflecting the rainfall gradient from south to north.

The belts from south to north

- **Freshwater and mangrove swamp** — the coast: mangroves in brackish water, palms, raffia.
- **Tropical rainforest** — 1500 mm+ of rain all year: tall, evergreen, layered forest with mahogany, obeche, iroko, and dense undergrowth.
- **Guinea savanna** — tall grasses with scattered trees (locust bean, shea, baobab); fires in the dry season.
- **Sudan savanna** — shorter grasses, thorny shrubs, acacia; the cattle country.
- **Sahel (semi-desert)** — very sparse grasses and acacia, bordering the Sahara.

Factors and threats

- Rainfall is the chief factor; soil and man also matter. **Deforestation** (logging, farming, fuelwood) and **overgrazing** + bush burning cause desertification — the southward advance of the desert in the Sahel.

CLASS EVALUATION

1. Name four vegetation belts of West Africa in order.
2. Which factors determine vegetation?
3. State two effects of deforestation.
4. What is desertification?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS1

SUBJECT: Geography

TOPIC: Map Reading & Scale

Introduction

- Topographical maps use conventional signs and scale to show relief, drainage, vegetation, settlement and land use. Map reading is the interpretation of these features.

Types of scale

- **Statement scale** — '1 cm represents 500 m'.
- **Representative fraction (R.F.)** — 1:50,000 (1 cm on the map = 50,000 cm = 500 m on the ground).
- **Linear scale** — a graduated line with distances marked; measure with a ruler or dividers and read directly from the line.

Calculations

- Ground distance = measured map distance × the scale factor (convert to km); area uses the square of the scale. Example: 6 cm on a 1:50,000 map = $6 \times 500 \text{ m} = 3 \text{ km}$.

Interpreting features

- Contours close together mean a steep slope; widely spaced – a gentle slope; V-shaped contours pointing uphill – a valley; concentric rings – a hill. Blue = water, green = low vegetation/planned areas, brown = relief, red = roads.

CLASS EVALUATION

1. Convert 1:50,000 to a statement scale.
2. What is the ground distance of 4 cm on a 1:25,000 map?
3. How do contours show a steep slope?
4. State the map colour for water and for roads.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS1

SUBJECT: Geography

**TOPIC: Drawing & Interpreting Simple Diagrams
(Cross-Sections)**

Introduction

- A cross-section is a profile of the land along a line drawn on the map; it shows how the relief rises and falls. Drawing and describing cross-sections is a standard practical skill.

Steps in drawing a cross-section

- Lay a strip of paper along the given line; mark every contour crossing with its height; transfer the marks to graph paper using a suitable vertical scale (e.g. 1 cm = 50 m) — exaggerating the vertical scale makes slopes visible; join the points with a smooth curve; label the vertical and horizontal scales and the features on the profile.

Describing a profile

- Use the correct vocabulary: steep/gentle slope, plateau, valley, ridge, escarpment, river line, settlement sites. Note the gradient between two points = vertical interval ÷ horizontal distance, and whether the route is suited to a road or railway.

CLASS EVALUATION

1. What is a cross-section?
2. Why is the vertical scale usually exaggerated?
3. State the difference between a ridge and a valley on a profile.
4. How do you calculate the gradient between two points?

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS1

SUBJECT: Geography

TOPIC: The Weather Station (Practical Application)

Introduction

- A weather station is a site where weather elements are observed and recorded daily with standard instruments. The school weather station gives practical experience and data for describing the local climate.

Setting up a weather station

- A level, open site away from tall buildings and trees; a Stevenson screen on a stand with its door facing north (to avoid direct sun); the rain gauge planted with its rim 30 cm above the ground; the barometer indoors; the anemometer and wind vane on a mast 10 m high; and a record sheet with the daily readings, taken at the same time each morning.

Recording and reading the data

- Record maximum and minimum temperatures; the rainfall total; wind speed and direction; humidity and cloud cover. From the data, compute the daily mean, the diurnal range, the monthly total, and the annual rainfall; draw a climate graph (bar chart of rainfall + line graph of temperature) and describe the climate: hot and wet all year — equatorial; two seasons — tropical; dry summers — Mediterranean.

CLASS EVALUATION

1. List four instruments in a weather station and what they measure.
2. Why must the Stevenson screen face away from direct sun?
3. How is the daily mean temperature calculated?
4. Describe the climate shown by high temperature and heavy rain all year.